

ON-DEVICE SLM × SAS/R • STUDY CP-101

A morning of on-device triage

Numbers from SAS/R, words from a small local model.

A scheduled SAS/R job runs the deterministic checks and Pinnacle 21 produces the conformance findings — exactly as today. A small open-weight model, running offline on the workstation, only labels each finding, sorts it into a queue, and drafts a candidate query. SAS/R owns every number; the model just helps with words; a human approves before anything is written.

Illustrative mockups. The model never computes or changes a number; validated tools and SAS/R own every value.

Scheduler

nightly batch



SAS / R + Pinnacle 21

every check & count



Local SLM (offline)

label · route · draft — words only



Validator + human

allowlist · approve



Write + Part 11 audit

model digest pinned

An ordinary workstation, Ollama, one pinned model — offline

workstation • terminal (offline)

```
$ ollama serve                # bound to 127.0.0.1:11434
$ ollama pull qwen2.5:7b-instruct-q8_0
  pulled · sha256 a1f3...9c  (Apache-2.0)

PINNED:
model    qwen2.5:7b-instruct-q8_0
digest   sha256:a1f3...9c
decode   temperature 0 · seed 42
ctx       8192 · batch 1

$ curl 127.0.0.1:11434/api/version → 200 OK
```

◇ No GPU farm. No cloud account. No telemetry.

A 7B model at Q8 runs on the workstation you already own. The same input gives the same output every run — the model, quantization, digest, and decoding are pinned.

≡ OFFLINE

The network cable is unplugged. SAS/R only ever calls 127.0.0.1 — and asserts the loopback before every run. The data-egress story is simply: nothing can leave.

Overnight, SAS/R + Pinnacle 21 run every deterministic check

batch log • validated work

```
[02:00] Pinnacle 21 CDISC conformance ...  
[02:14] SDTM edit-checks (cross-form) ...  
[02:21] PK timing / BLQ rules ...  
[02:28] lab / IXRS reconciliation ...
```

FINDINGS: 214

Error 38 • Warning 121 • Notice 55

```
# the small model has NOT been called.  
# every count came from validated code.
```

findings (a sample) — owned by SAS/R

ID	Rule	Finding (short)
SD0064	AE date order	AESTDTC after AEENDTC ×3 (AE)
SD0011	Required var	LBSTRESN missing ×12 (num)
PK0003	Timing dev	ADPC 4h actual >10% off nominal ×2
SD0204	CT term	Non-CT value in AEACN ×1
RC0007	Lab recon	3 LB records unmatched to transfer

Counts, rules, comparisons, reconciliation — all from validated, double-programmed code.

SAS/R sends each short finding for three narrow calls — JSON only

⋮ SAS • PROC HTTP → 127.0.0.1/api/chat

```
%slm_init(model=&CFG_MODEL,
          base=http://127.0.0.1:11434);
↳ loopback asserted • model pinned

%slm_classify(
  usr=%nrstr(AESTDTC is after AEENDTC
             for 3 records in AE),
  field=owner, allow=&CFG_OWNERS);

format = JSON SCHEMA (enum-constrained)
options: temperature 0 • seed 42 • stream off
```

⋮ the model's reply — schema-constrained JSON

```
{
  "novelty": "NEW",
  "owner": "PROGRAMMING",
  "severity": "MAJOR",
  "confidence": 0.86,
  "rationale": "start after end
               date – likely keying"
}
```

Short-text classification + routing into a fixed vocabulary — squarely what a small quantized model does reliably. No prose, no numbers.

SAS/R validates every field, then drafts a grounded query

⋮ SAS • %slm_validate (the trust boundary)

```

parse JSON ..... ok
owner in CFG_OWNERS ..... ok
severity in CFG_SEVERITY ... ok
confidence 0-1 ..... ok

# any off-allowlist value →
REJECT + log, never coerce.
# 211 valid • 3 rejected (re-queued)

draft query: grounded by RAG over
the study's own spec + edit-check defs

```

⋮ draft query — grounded, marked DRAFT, unsent

DRAFT • not sent — SD0064 • AE

Please verify the AE onset/resolution dates:
 AESTDTC (start) is recorded after AEENDTC
 (end) for 3 records. Per the edit-check spec
 (EC-AE-07), confirm and correct the dates or
 confirm the event sequence.

Wording stays anchored to the study's real spec text — not the model's priors.

The biostatistician works a ranked worklist — facts beside suggestions

⋮ Triage worklist — SAS/R facts • model suggestion • your decision

ID	SAS/R facts	Novelty	Owner	Sev	Draft query	Decision
SD0064	AESTDTC>AEENDTC ×3 (AE)	NEW	PROGRAMMING	MAJOR	verify AE dates...	✓ Approve
RC0007	3 LB unmatched to transfer	NEW	DM	MAJOR	reconcile LB...	✓ Approve
PK0003	ADPC 4h >10% off ×2	KNOWN	PK	MINOR	confirm timing dev...	✎ Edit
SD0011	LBSTRESN missing ×12	NEW	DM	MAJOR	query missing num...	✓ Approve
SD0204	Non-CT value AEACN ×1	KNOWN	MEDICAL CODING	MINOR	(expected — explained)	✗ Reject

Each row shows the deterministic facts from SAS/R next to the model's label and draft. A wrong label is a two-second fix — nothing becomes real until a qualified human signs off.

6 The model triaged; the biostatistician decides. Approve, edit, or reject each one — the human gate is the point at which a suggestion can become an action.

Approved items are written — with the model digest pinned

audit/triage_audit.txt — one line per approved item

```
2026-06-16T07:42 SD0064
  owner=PROGRAMMING novelty=NEW sev=MAJOR
  model=qwen2.5:7b-instruct-q8_0
  digest=sha256:alf3...9c seed=42 temp=0
  draft="verify AE dates..."
  reviewer=pkiben decision=APPROVED
  query_id=Q-10421 (issued to EDC)

2026-06-16T07:43 RC0007 ...APPROVED
2026-06-16T07:44 SD0204 ...REJECTED (no write)
```

Re-runnable for an inspector

Because the model is frozen and decoding is fixed, the audit line is a reproducible artifact: same input, same model digest, same output.

Offline — proven

The cable is unplugged. The only thing that ever left the box was a query you approved — into your own EDC.

The honest shape of an on-device win

SAS / R + Pinnacle 21

- All data access & computation
- Every check, count & reconciliation
- The output validator (allowlist)
- Every write + the Part 11 audit

The local small model

- Labelled new-vs-known
- Classified owner & severity
- Drafted a grounded candidate query
- — and nothing else; no numbers

The biostatistician

- Read the ranked worklist
- Approved, edited, or rejected
- Owned every decision that mattered
- Signed off before any write

The model saved triage time on a narrow, validated task — and the truth stayed exactly where it always was: with SAS/R, the validated engines, and a qualified human.

Illustrative. On-device, offline, no number ever produced by the model.